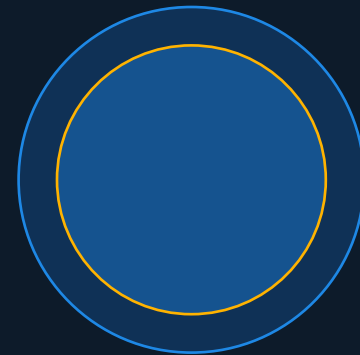


AIRLINE CUSTOMER SATISFACTION ANALYSIS



*Predicting What Drives Passenger Satisfaction
Using Machine Learning on 129,880 Records*

129,880

Passengers

22

Features

54.7%

Satisfied

3 Models

Tested

PROJECT OVERVIEW

Business Problem

Airlines face high competition where passenger experience defines loyalty. This project identifies which service attributes most influence satisfaction to help airlines prioritize improvements and reduce churn.

Objective

Build a predictive model that classifies passengers as Satisfied or Dissatisfied, and uncovers the top drivers of satisfaction.

Analytical Approach

- Data Cleaning & Duplicate Removal
- Missing Value Treatment (Median)
- Outlier Treatment (Capping Method)
- Feature Selection: ANOVA & Chi-Square
- ML Models: Logistic Reg + Decision Tree
- 10-Fold Cross Validation

DATASET AT A GLANCE

129,880

Total Records

No duplicates found

22

Features

21 predictors + 1 target

54.7%

Satisfied

71,087 passengers

393

Missing Values

Arrival Delay only → Median filled

Feature Categories

Demographics

Age (7–85 yrs)
Customer Type
(Loyal / Disloyal)

Trip Info

Travel Type
(Business / Personal)
Class, Flight Distance

Service Ratings

14 service attributes
rated 1–5
(Seat comfort, WiFi...)

Operational

Departure Delay
Arrival Delay
(minutes)

WHO IS SATISFIED? PASSENGER SEGMENTATION

★ **KEY INSIGHT:** Business travellers in Business class are the most satisfied segment. Personal travellers are largely dissatisfied regardless of class.

Satisfaction Rate by Customer & Travel Segment

Loyal Customer – Business Travel

78%

22%

Loyal Customer – Personal Travel

41%

59%

Disloyal Customer – Business Travel

28%

72%

Disloyal Customer – Personal Travel

8
%

92%

Business Class: ~69% satisfied | Eco/Eco Plus: ~44% satisfied



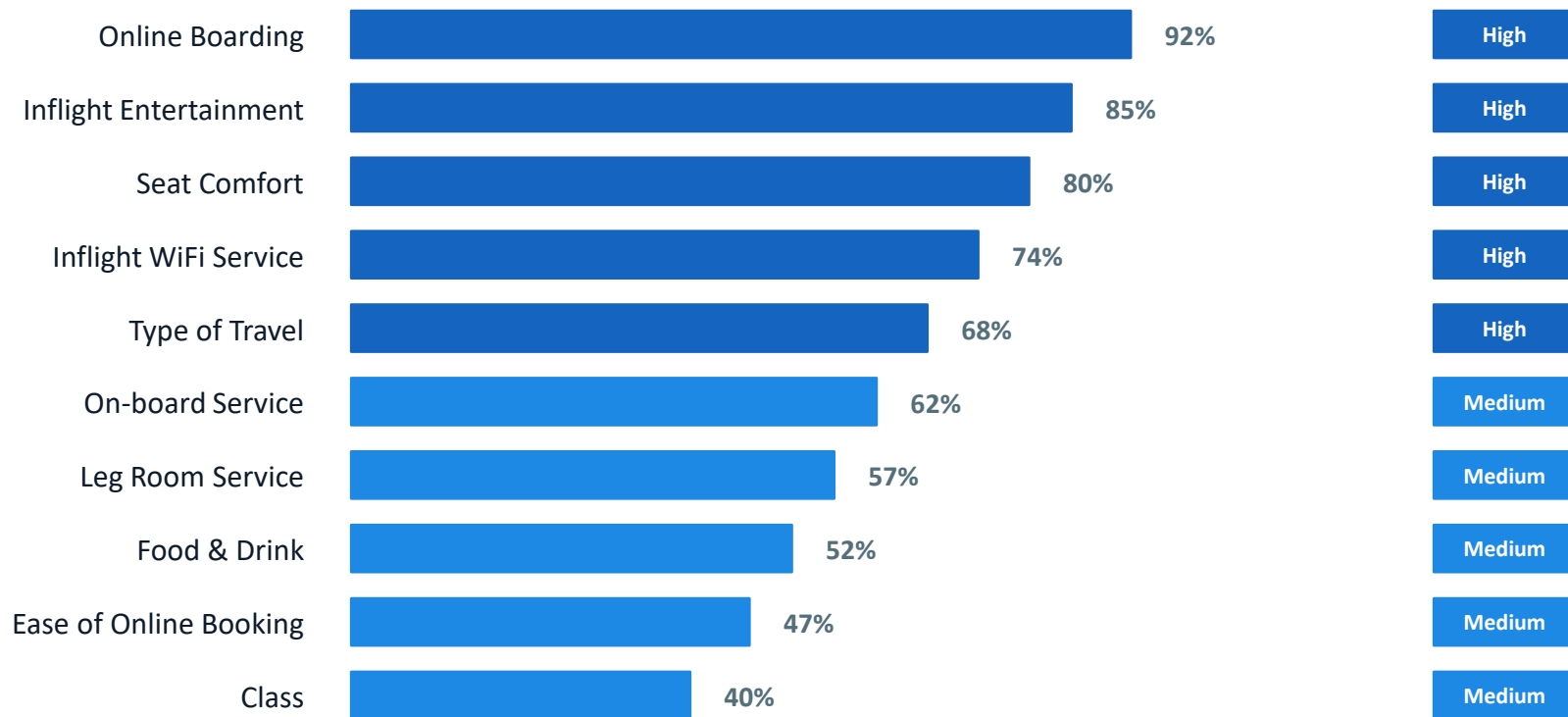
Satisfied



Dissatisfied

TOP DRIVERS OF SATISFACTION

All 21 features passed Chi-Square & ANOVA tests ($p < 0.05$) — confirming strong correlation with satisfaction



LOYALTY & TRAVEL TYPE — CRITICAL DIFFERENTIATORS

81.7%

Loyal Customers

106,100 of 129,880 passengers

61.5% satisfaction rate
Loyals are 5.6x more likely to be satisfied than disloyals

18.3%

Disloyal Customers

23,780 passengers

23.9% satisfaction rate
High churn risk — dissatisfaction is dominant

Travel Type Split

Business Travel

89,693 passengers (69.1%)

Satisfaction Rate: ~71%

Personal Travel

40,187 passengers (30.9%)

Satisfaction Rate: ~24%

★ Business travellers are 3x more satisfied than personal travellers

SERVICE QUALITY — WHERE GAPS EXIST

★ Strengths

Baggage Handling

Consistently rated well across all classes

On-board Service

Strong in business class passengers

Checkin Service

Low friction — high satisfaction

Inflight Entertainment

Key differentiator for business travellers

⚠ Areas for Improvement

Inflight WiFi

Top complaint — low ratings across all segments

Ease of Online Booking

Significant dissatisfaction in personal travellers

Departure/Arrival Timing

Weakest predictor — scheduling constraints

Gate Location

Inconvenience noted especially in Eco class

MACHINE LEARNING MODEL RESULTS

Logistic Regression

83%

Test Accuracy

10-Fold CV:

77%

Avg CV Score

Class-wise Performance:

Class	Precision	Recall
Dissatisfied	81%	81%
Satisfied	84%	84%

Stable but lower cross-validation score — suggests some variability

Decision Tree ★ BEST

85%

Test Accuracy

10-Fold CV:

80%

Avg CV Score

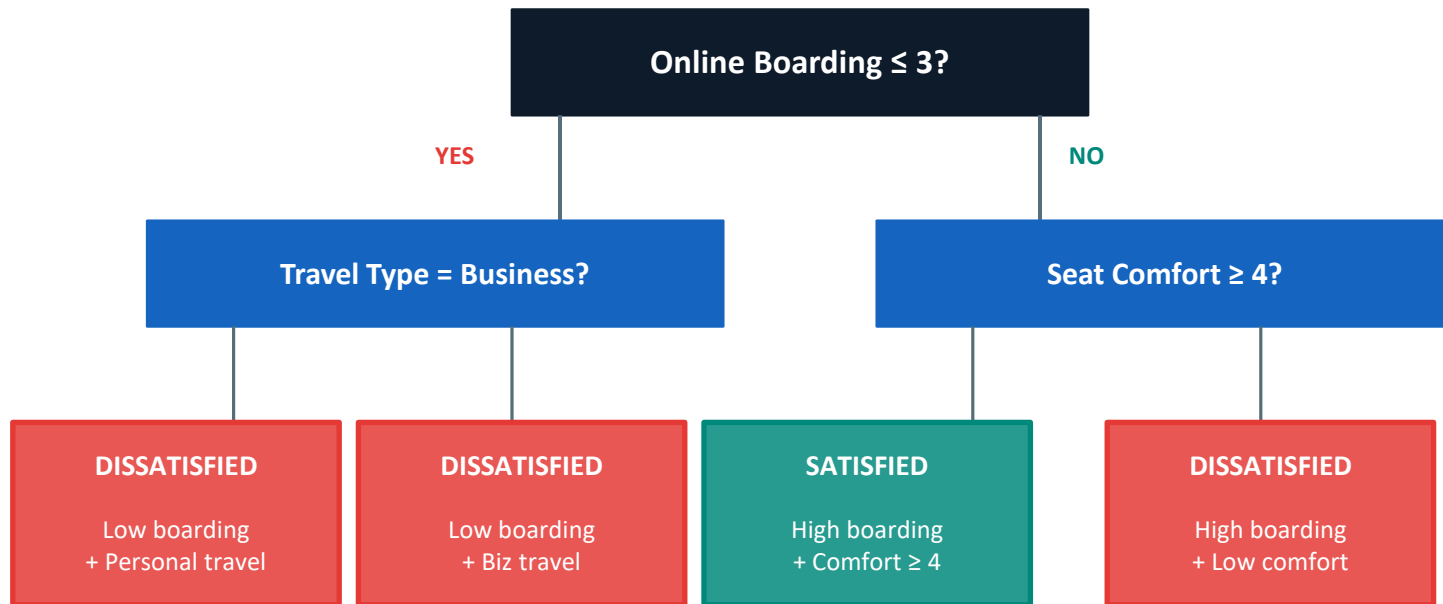
Class-wise Performance:

Class	Precision	Recall
Dissatisfied	90%	74%
Satisfied	81%	93%

Better overall accuracy with interpretable rules — preferred model

DECISION TREE: HOW THE MODEL DECIDES

The Decision Tree (depth=3, entropy criterion) reveals the top predictors through clear IF-THEN rules:



Online Boarding quality is the #1 split point — passengers who rate it ≤ 3 are overwhelmingly dissatisfied. Seat Comfort is the key secondary driver for those who board well.

KEY RECOMMENDATIONS

01

Fix Online Boarding Experience

The #1 decision tree split. Invest in app UX, boarding gate notifications, and streamlined check-in to directly lift satisfaction scores.

02

Upgrade WiFi & Inflight Entertainment

Both rank in the top 5 satisfaction drivers. Poor WiFi is a top complaint — reliable connectivity is now a baseline expectation.

03

Prioritize Loyal Customer Retention

Loyal customers drive 81.7% of volume with 61.5% satisfaction. A 5% improvement in loyalty satisfaction could transform revenue.

04

Target Personal Travellers with Value

Only 24% of personal travellers are satisfied. Bundled perks (lounge access, seat upgrades) could convert this high-volume segment.